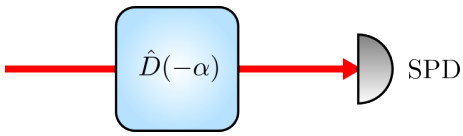


Figure 3.15: Displaced single-photon detection. An optical displacement by $-\alpha$ followed by a bucket single-photon detector (SPD) implements a projective measurement onto the coherent state $|\alpha\rangle$.

### Homodyne detection

Photodetection also enables measurements of continuous degrees of freedom of light, in particular of the quadrature operators $\hat{x}$ and $\hat{p}$ (or more generally along any direction in phase space $\hat{x}(\theta)$). Quadrature measurements can be realised with *homodyne detection*, sometimes just called homodyning. Homodyning is also used in classical signal processing e.g. in radio technology. For quantum light, homodyne detection can be implemented as illustrated in Fig. 3.16. The mode to be measured is mixed on a balanced beam splitter with a bright coherent state (i.e. a bright laser beam), known as the *local oscillator*. Each output is sent to a photodetector that registers intensity. Finally, the difference between the signals is computed.

To see that this implements a quadrature measurement, consider an arbitrary (pure) input state $|\psi\rangle$ in the signal mode ($A$) and a coherent state $|\beta\rangle$ in the local-oscillator mode ($B$). The two detectors measure the photon numbers $\hat{n}_A = \hat{a}^\dagger\hat{a}$ and $\hat{n}_B = \hat{b}^\dagger\hat{b}$ (but without single-photon resolution) in the appropriate modes (corresponding to integrating the intensities over some detection interval). The expected output, after taking the difference, is then

$$\Delta I = {}_{AB}\langle\psi,\beta|U_{BS}^\dagger(\hat{a}^\dagger\hat{a} - \hat{b}^\dagger\hat{b})U_{BS}|\psi,\beta\rangle_{AB} \tag{3.167}$$

$$= \frac{1}{2}{}_{AB}\langle\psi,\beta|\left[(\hat{a}^\dagger + \hat{b}^\dagger)(\hat{a} + \hat{b}) - (\hat{a}^\dagger - \hat{b}^\dagger)(\hat{a} - \hat{b})\right]|\psi,\beta\rangle_{AB} \tag{3.168}$$

$$= {}_{AB}\langle\psi,\beta|\left[\hat{a}^\dagger\hat{b} + \hat{a}\hat{b}^\dagger\right]|\psi,\beta\rangle_{AB} \tag{3.169}$$

$$= {}_{A}\langle\psi|\left[\hat{a}^\dagger\beta + \hat{a}\beta^*\right]|\psi\rangle_{A} \tag{3.170}$$

where, in the second line we have used the inverse of the beam-splitter transformation (3.134)-(3.135), and in the fourth line we have used that ${}_B\langle\beta|\hat{b}|\beta\rangle_B = \beta$ and ${}_B\langle\beta|\hat{b}^\dagger|\beta\rangle_B = \beta^*$. Let $\beta = |\beta|e^{i\theta}$. Then

$$\Delta I = |\beta|\,{}_A\langle\psi|\left[\cos(\theta)(\hat{a} + \hat{a}^\dagger) - i\sin(\theta)(\hat{a} - \hat{a}^\dagger)\right]|\psi\rangle_A \tag{3.171}$$

$$= \sqrt{2}|\beta|\,{}_A\langle\psi|\left[\cos(\theta)\hat{x}_A + \sin(\theta)\hat{p}_A\right]|\psi\rangle_A \tag{3.172}$$

$$= \sqrt{2}|\beta|\,{}_A\langle\psi|\hat{x}_A(\theta)|\psi\rangle_A = \sqrt{2}|\beta|\,\langle\hat{x}_A(\theta)\rangle_\psi, \tag{3.173}$$

where we have used (3.20)-(3.21) and (3.53). The expected output is thus proportional to the expectation value of the rotated quadrature $\hat{x}(\theta)$ of the input signal mode. Since this holds for every state $|\psi\rangle$, the measurement is equivalent to a projective measurement of $\hat{x}(\theta)$. Note that the signal is effectively being amplified by a factor of $|\beta|$ (the square root of the mean photon number in the local oscillator), which can be very large. We can choose the direction

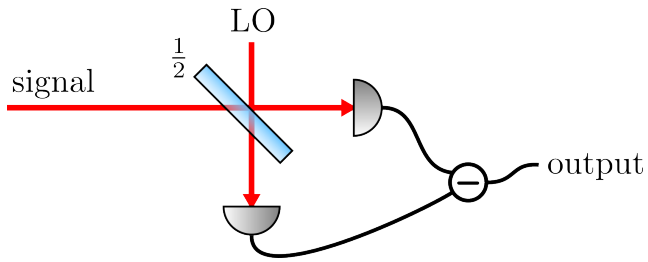


Figure 3.16: Homodyne detection. The mode to be measured (signal) is mixed on a balanced beam splitter with a a coherent state, the local oscillator (LO). The intensity at each output port is detected and the difference of the results is computed.

of the measured quadrature by adjusting the phase $\theta$ of the local oscillator. In particular, for $\theta = 0$ and $\theta = \frac{\pi}{2}$, measurements of $\hat{x}$ and $\hat{p}$ are realised, respectively.

Because it does not require single-photon detectors and can be implemented with highly efficient photodiodes, homodyning can be realised with very high efficiencies ($\geq 99\%$) in practice.

### Heterodyne detection

Homodyning enables projective measurements of a single quadrature. Although conjugate quadratures such as $\hat{x}$ and $\hat{p}$ do not commute and hence cannot be measured perfectly at the same time, it is nevertheless sometimes desirable to perform a noisy simultaneous measurement of them. This can be done using *heterodyne detection*.

Heterodyne detection can be implemented in a similar setup as homodyne detection (Fig. 3.16) by shifting the frequency of the local oscillator slightly relative to that of the signal and performing some electronic filtering of the detector outputs. This use of different frequencies is the common meaning of the term heterodyne detection in classical signal processing and it also works in the quantum regime. However, it is equivalent to a setup using a single frequency and two homodyne detectors, which we present here, as it is somewhat simpler to describe. The scheme is illustrated in Fig. 3.17.

To realise a simultaneous measurement of $\hat{x}$ and $\hat{p}$, the signal mode ($A$) is first split on a balanced beam splitter, with vacuum entering the other input port ($B$). Each output mode is then measured using homodyne detection, with different phases of the local oscillators corresponding to detecting $\hat{x}$ and $\hat{p}$. Because of the first balanced beam splitter, the measured quadratures correspond to

$$\frac{1}{\sqrt{2}}(\hat{x}_A + \hat{x}_B) \tag{3.174}$$

and

$$\frac{1}{\sqrt{2}}(\hat{p}_A - \hat{p}_B). \tag{3.175}$$

Since mode $B$ is in the vacuum state, $\langle \hat{x}_B \rangle = \langle \hat{p}_B \rangle = 0$ and the expectation values of the homodyne outputs are proportional to $\langle \hat{x}_A \rangle_\psi$ and $\langle \hat{x}_A \rangle_\psi$, where